MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, FIFTEEN INDIVIDUALLY SHIELDED PAIRS OF 22 AWG SOLID WIRE

NOTE: NOT FOR AEROSPACE USE.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49285.

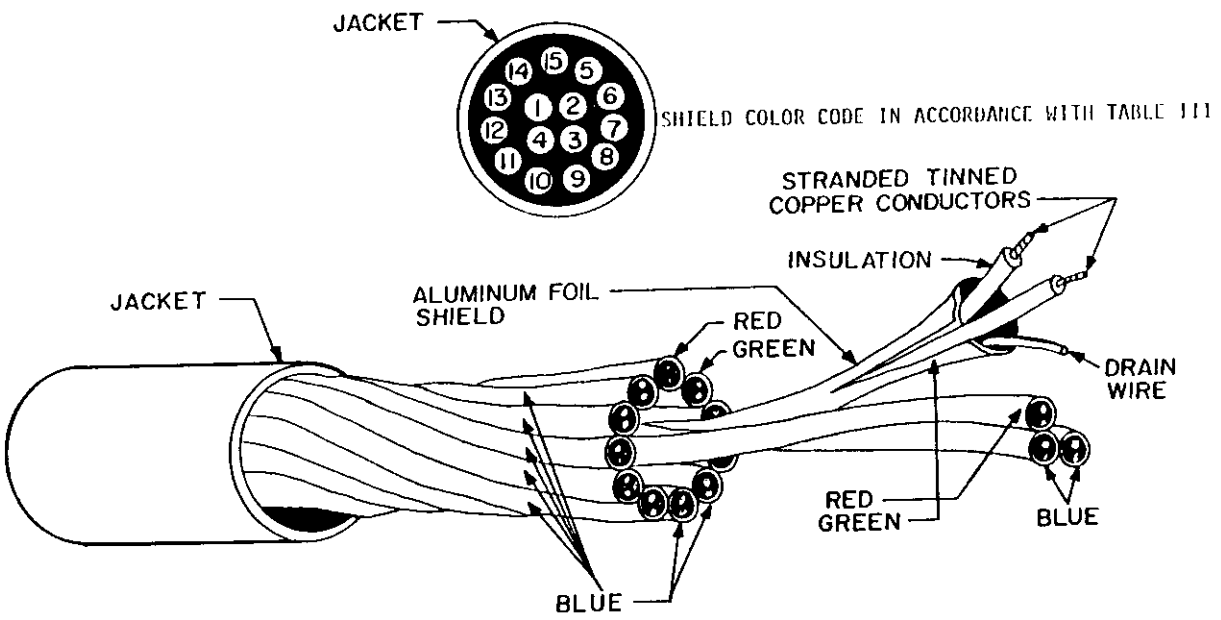


FIGURE 1. Cable illustration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Engineering information: (See table I).

TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	40 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	77 pF/ft
Continuous working voltage	300 V rms maximum between conductors
Nominal capacitance between adjacent shields	150 pF/ft
Continuous working voltage between adjacent shields	50 volts maximum
Shield dc resistance	14 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of pairs	Fifteen individually shielded
Conductor type and wire size	Solid tinned copper 22 AWG
Conductor insulation	Polyvinyl chloride
Conductor insulation thickness	0.013 inch nominal thickness
Drain wire type and size	Solid tinned copper 22 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.044 inch
Finished cable diameter	0.551 inch maximum outside diameter
Cable style (UL)	Not applicable
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay lengths	1.3 twists per foot ±10 percent

Cabled pair orientation: The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1).

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TABLE III. Shield color code.

Pair numbers	Shield color
Center core	
1st pair	Red
2nd pair	Green
3rd pair	Blue - 1
4th pair	Blue - 2
Outer layer	
5th pair	Red
6th pair	Green
7th pair	Blue - 1
8th pair	Blue - 2
9th pair	Blue - 3
10th pair	Blue - 4
11th pair	Blue - 5
12th pair	Blue - 6
13th pair	Blue - 7
14th pair	Blue - 8
15th pair	Blue - 9

REQUIREMENTS:

Design and construction: (See tables II and III).

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair with the aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact.

Drain wire location. Each drain wire is to be spirally located beneath the shield with the insulated conductors. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 1581 Vertical-Tray flame test requirements.

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Shield integrity test.*

Frequency	Response **
100 kHz	114 dB below reference
500 kHz	105 dB below reference
1 MHz	99 dB below reference
5 MHz	85 dB below reference
10 MHz	79 dB below reference
15 MHz	74 dB below reference
20 MHz	70 dB below reference
25 MHz	65 dB below reference
30 MHz	61 dB below reference
100 MHz	42 dB below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Fixture resonance will occur between 40 and 70 MHz.

Crosstalk test limits. All values are minimums.

Frequency (kHz)	Near end crosstalk (dB)	Far end crosstalk (dB)
40	58	58
70	56	55
100	54	53
1,000	55	52

Insulation resistance between shields: 100 megohms/1,000 feet (minimum).

Durometer hardness. The cable jacket shall have a "Shore A" hardness of 82 ±5.

Part or Identifying Number (PIN): The PIN shall be M49285/17.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Air Force - 71
DLA - ES, IS

Preparing activity:
Army - CR

Agent:
DLA - ES

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